TEAM 16

ENHANCING THE MUSIC FESTIVAL EXPERIENCE WITH

PULSEBAND



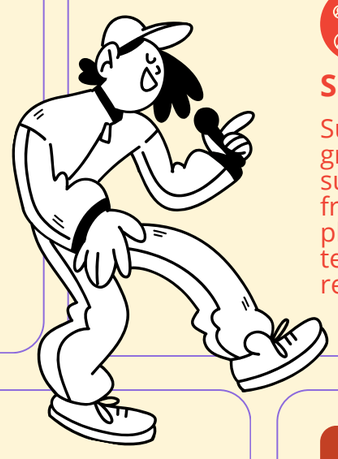




A SOCIAL WEARABLE CONCEPT THAT HELPS FRIENDS COMMUNICATE, STAY AWARE OF EACH OTHER, AND RECONNECT DURING CONCERTS AND MUSIC FESTIVALS WITHOUT CONSTANTLY RELYING ON THEIR PHONES.

PROBLEM SPACE

Concerts and music festivals often make it difficult for friends to stay connected. Poor phone reception, loud music, and crowded environments can prevent communication and make finding separated friends challenging. As a result, attendees may miss performances, feel unsafe, or spend time searching for their group instead of enjoying the event.



DESIGN OPPORTUNITY

"How might we support friends in staying connected and communicating changing plans without distracting them from the live experience?"

SOCIAL

Supports existing group behaviours such as checking on friends, sharing plans, separating temporarily, and reconnecting.

MOBILE

Responds to changing movement, proximity, noise and connectivity as people move between stages, toilets, food areas and crowds.

TARGET AUDIENCE

Young adults who attend concerts and music festivals with friends, particularly at larger events where groups are likely to move or separate.

RESEARCH EVIDENCE

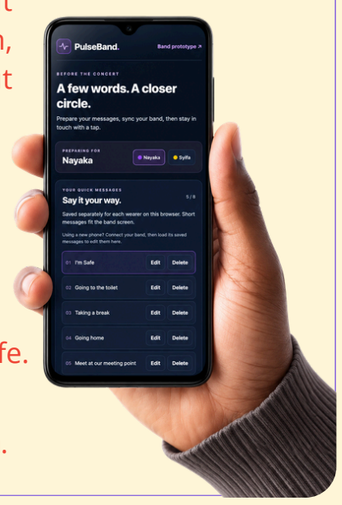
SURVEYS
Identified broader patterns in event behaviour, communication, and common frustrations.

INTERVIEWS
Explored separation, mobile reception, crowding, phone use, accessibility, and existing coping strategies.

USER TESTS
Evaluated friend-finding, proximity feedback, directional guidance, and interface clarity.

OUR CONCEPT

PulseBand is a social wearable for small groups of friends at live music events. It enables quick, simple communication, and helps users reconnect without typing on their phones.



FIND FRIEND
Directional and proximity feedback to reconnect.

QUICK SIGNALS
Send simple updates such as Toilet, Taking a Break, Going Home, and I'm Safe.

SOS
Send an urgent signal with confirmation.

WHAT WE LEARNED FROM USERS



1. UNRELIABLE CONNECTIVITY

Poor mobile reception and loud environments make calls and messages unreliable, especially when friends become separated.



2. DIFFICULT RECONNECTION

Participants currently use meeting points, landmarks, messaging apps, waiting, and visual searching to find each other but these methods can fail when crowds are moving.



3. EXPERIENCE DISRUPTION

Searching for friends and repeatedly checking phones can distract attendees from performances and increase safety concerns.

CONCEPT DEVELOPMENT

1. **EARLY EXPLORATION**
Explored different wearable forms, navigation methods, and interaction possibilities.

2. **RESEARCH AND FOCUS**
Interviews, surveys and affinity mapping highlighted friend seperation, poor reception and communication as key needs.

3. **LOW AND MID-FIDELITY PROTOTYPING**
created a sketches and developed a mid-fidelity interface to explore how users navigate between features, send pre-made messages, and find their friends.

4. **HIGH-FIDELITY PROTOTYPING**
Built a wearable prototype with directional arrows, pre-made messaging, and an SOS button, alongside a website for customising and syncing messages to the wearable.

USER TESTING AND DESIGN REFINEMENT

WHAT WORKED

USEFUL FOR RECONNECTION
Participants could see how the concept could help when friends become separated.

"We go to concerts quite a lot, so I think something like this would be really helpful."

DIRECTIONAL GUIDANCE WAS VALUABLE
Participants liked having a clear cue showing where to move.

"Especially the arrow thing. I think that's really useful because you can just follow it."



WHAT NEEDED REFINEMENT

1. ARROW CONSISTENCY
Worked when rotating, but became inaccurate during sideways movement.

"When I tried sliding it sideways, it kind of went off in the wrong direction."

2. PROXIMITY LIGHT VISIBILITY
Brightness changes were too subtle.

"Probably not if it's just from the brightness, no."

3. REDUNDANT PROXIMITY DOTS
Participants did not understand the purpose of the four proximity dots.

"I didn't realise they actually meant anything."
"I think the lights already kind of do that, though."

FINAL DESIGN OUTCOME

CLEARER DIRECTION
Improving arrow behaviour for more consistent guidance.

STRONGER PROXIMITY FEEDBACK
Replacing subtle brightness-only changes with more noticeable blinking as friends get closer.

LESS VISUAL CLUTTER
Removing proximity dots that participants found unclear and redundant.

FOCUSED INTERACTION
Keeping the experience centred on Find Friend, quick signals, and SOS rather than adding unnecessary features.

STAY CONNECTED. FEEL THE MOMENT.

PulseBand supports existing social behaviours rather than replacing them. Friends remain responsible for deciding when to separate, communicate, reconnect, or respond. The wearable simply provides lightweight awareness when that support is needed.